

RESEARCH REPORT

The Test-Retest and Interrater Reliability of the Promoting the Emergence of Advanced Knowledge-Direct Training Assessment for Use With Individuals With Autism and Related Disabilities

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The study evaluated the test-retest and interrater reliability of the Promoting the Emergence of Advanced Knowledge-Direct Training (PEAK-DT) assessment. In the study, the reliability of PEAK-DT scores were assessed both in terms of raw scores and age-referenced scores. To evaluate test-retest reliability, 39 participants with autism or a related disability were administered the PEAK-DT assessment on 2 occasions—during an initial assessment and a follow-up assessment after 3 weeks to 1 month. The results suggest that PEAK-DT had high test-retest reliability both for raw scores (intraclass correlation coefficient [ICC] = .987, $p < .001$) and age-referenced scores (ICC = .973, $p < .001$). In addition, 10 of the participants were assessed independently by 2 observers during the initial assessment period to evaluate interrater reliability. The results also suggested that PEAK-DT had strong interrater reliability ($\kappa = .981$, $p < .001$). Taken together, the results provide support for the reliability of PEAK-DT as a behavioral assessment of directly trained language skills.

Keywords: ABA therapy, autism assessment and treatment, relational frame theory, PEAK relational training system

The number of individuals with an intellectual or developmental disability has increased over the past decade, with one study reporting that 1 in 6 children have a diagnosed developmental or intellectual disability (Boyle et al., 2011). Those with a developmental or intellectual disability usually exhibit some deficit in intellectual functioning or adaptive functioning, especially in terms of language development (American Psychiatric Association, 2013). As a result, many applied behavior analytic (ABA) based treatments and interventions have emerged as a means of facilitating the develop-

ment of language to counteract the deficits experienced by individuals in these populations. Virués-Ortega (2010) reports that the use of long-term ABA treatments can result in significant gains in IQ, adaptive behavior, and language. Although ABA-based treatments are effective, clinicians need to be able to identify exactly which skills are lacking in an individual's language repertoire. Several assessment guides have been developed that provide clinicians with such a tool; unfortunately, little to no data has been published supporting the psychometric utility of many existing behavior analytic assessments of language. Without psychometric support, the validity of the assessments and the reliability of the obtained data cannot be assured.

The Promoting the Emergence of Advanced Knowledge Relational Training System (PEAK, Dixon, 2014) is one assessment that is designed to identify specific language deficits experienced by special populations in order to guide treatment decisions. Recently, data has been generated supporting the validity of the

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PEAK-DT assessment. For example, [Dixon, Whiting, Rowsey, and Belisle \(2014\)](#) demonstrated that PEAK-DT total scores are correlated with several measures of IQ for individuals with disabilities. In addition, scores on the PEAK-DT assessment have been shown to relate to vocabulary ([Dixon et al., 2015](#)) and other measures of language from a behavioral paradigm ([Dixon, Carman, et al., 2014](#)). While ensuring the validity of data is important, those measures are of no value without assurance that the data are reliable. One form of reliability that has not been evaluated for PEAK -DT is the test-retest reliability of the assessment. Test-retest reliability is the extent to which an assessment produces similar results across repeated tests of the same individual. For an assessment to be reliable, there is an assumption that the construct being measured is also stable. Test-retest reliability assesses the degree to which a score at one time is predictive of a score following the passage of some amount of time. Establishing the stability of scores on a reliable assessment when targeted interventions are absent (i.e., in a baseline condition) is necessary to then evaluate how a treatment or intervention influences that measure over time. Often, determining the ICC ([Bartko, 1966](#)) between two time periods provides a measure of the test-retest reliability of an instrument ([Somandepalli et al., 2015](#)). Correlation coefficients of a range from 0.0 to 1.0, and a score above 0.7 is generally considered indicative of a reliable assessment provided those results are statistically significant (i.e., unlikely to have occurred by chance alone). Another form of reliability that may require further empirical exploration for PEAK-DT is interrater reliability. Several studies have used interobserver agreement (IOA) to support the reliability of the PEAK-DT assessment ([Dixon, Belisle, Whiting, & Rowsey, 2014](#); [Dixon, Carmen, et al., 2014](#)), but a more systematic evaluation is required. IOA can provide a biased score when data are dichotomous (i.e., PEAK-DT), and may present biased accounts of reliability due to chance occurrence. [Cooper, Heron, and Heward \(2007\)](#) suggest that assessing IOA using only scored data, or only on intervals where at least one observer recorded an occurrence of behavior, may reduce a bias associated with low scores. Another way to evaluate interrater reliability that reduces bias is using Cohen's kappa ([Cohen, 1965](#)) as a mea-

sure of reliability, which allows for the elimination of agreement that is likely to occur by chance and deals specifically with dichotomous data. At present, these two measures of reliability need to be assessed to provide a better understanding of the PEAK-DT assessment.

Another consideration is that PEAK scores are often reported as a raw score, which may only be informative to individuals with experience conducting PEAK, and may reduce the translational utility of the score to parents and caregivers with less familiarity with the assessment. Two studies have been conducted that provide a method for reporting age-referenced PEAK scores in addition to raw PEAK scores. The first was conducted by [Rowsey, Belisle, and Dixon \(2015\)](#), which examined the underlying constructs of the PEAK-DT assessment. The component analysis indicated that PEAK-DT consists of four components: Foundational Learning Skills; Perceptual Learning Skills; Verbal Comprehension Skills; and Verbal Reasoning, Memory, and Mathematical Skills. Foundational learning skills assess basic instruction following and attending to the environment; perceptual learning skills assess basic environmental interactions and cognitive abilities; verbal comprehension skills assess more complex linguistic and cognitive abilities; and verbal reasoning, memory, and mathematic skills measure basic logic processes and advanced cognitive abilities needed for effective social behavior. The second study, conducted by [Dixon, Belisle, Whiting, et al. \(2014\)](#), developed a normative sample of PEAK-DT scores for comparison to individuals with autism. By synthesizing the results of these two studies, a clinician has the ability to report both raw scores and age-referenced scores across each component and for PEAK total scores. As with the PEAK-DT raw scores, the reliability of the PEAK-DT age-referenced scores must also be assessed to determine the reliability of the results garnered from the PEAK-DT assessment. Reporting the reliability of the age-referenced scores as well is important to ensure that the information provided to parents and caregivers is likely to be repeatable and stable across time and observers.

Therefore, the purpose of the present study was to provide an initial evaluation of the test-retest and interrater reliability of the PEAK-DT assessment. These forms of reliability were se-

lected because both analyses add to a psychometric understanding of the PEAK-DT assessment more generally, and because the combination of these two assessments allow for an understanding of how assessment scores are stable across multiple observers and across time. In addition, the reliability assessment was conducted with individuals with disabilities, to provide an indication of the reliability of the assessment with individuals who are most likely to benefit from the obtained results.

Method

Participants and Setting

The participant demographics are summarized in Table 1. A total of 39 participants (32 male, 7 female) between the ages of 6 and 22 ($\bar{X} = 12.9$, $s = 4.4$) took part in the study. A majority of the participants had an autism diagnosis ($n = 36$) and the remaining had a related intellectual or developmental disability ($n = 3$). All of the participants attended a self-contained school for individuals with autism. The PEAK-DT assessments were conducted by authors of the current study, who were familiar with the participants prior to the beginning of the study. Assessments were conducted either in the participant's classroom, or in a separate room in order to minimize distractions. Participants were intermittently provided with reinforcement to encourage participation. Access to the reinforcers were not contingent on correct or incorrect responses during the assessment, and were usually in the form of attention, a break from work, or a preferred edible.

Table 1
Demographic Information

Demographic	Group	<i>n</i> (%)
Gender	Male	31 (81.5)
	Female	7 (18.4)
Age	6–10	16 (41.0)
	11–15	11 (28.2)
	15–20	12 (30.7)
Diagnosis	Autism	36 (92.3)
	Other	3 (7.6)

PEAK-Direct Training Assessment

The PEAK-DT assessment contains 184 items that are used to assess an individual's directly trained skill repertoire. These skills can be assessed through either an indirect or direct observation approach. Items on the assessment are answered with a “yes,” “no,” or “?” response, to indicate whether or not a skill is in a participants' repertoire. If an assessor is unsure if the participant is able to successfully demonstrate a skill on the assessment, then the skill is directly tested. In the direct assessment, the assessor presents the relevant stimuli and delivers an instruction. If the participant is able to perform the skills specified in the instruction, then the assessor will indicate that the skill is in the participants' repertoire. The total number of items that the participant is able to demonstrate is totaled to arrive at the PEAK-DT Total score, which ranges from 0 to 184 (corresponding to the number of items in the assessment) and serves as composite measure of the participants' directly trained language repertoire. An assessor may also use the components identified by Rowsey et al. (2015) to indicate component scores, and the PEAK age-referenced table provided in the Appendix of the present paper can be used to indicate age-referenced scores for PEAK total scores as well as for the individual component scores.

Procedures and Data Analysis

Assessors began by conducting an indirect assessment of each participant using the PEAK-DT assessment. Assessors indicated that a participant was able to perform a certain item if they had observed the participant perform the skill in the past. If the assessor was unsure if a participant was able to perform a skill, then a “?” was indicated on the assessment. For all items where a “?” was indicated, the skill was directly tested.

To evaluate test–retest reliability, all of the participants were reevaluated 3 weeks to 1 month following their initial PEAK-DT assessment ($\bar{X} = 26.7$ days, $s = 4.2$ days). The interval between the initial assessment and the retest of the assessment was short enough to minimize confounds caused by the maturation or skill acquisition that the participants might exhibit over time, but long enough so that the assessors

would not recall the selections made on the initial test. Test–retest reliability was evaluated for both PEAK raw scores and PEAK age-referenced scores, across each PEAK component. Component scores were determined using the components identified by Rowsey, Belisle, and Dixon (2015). To evaluate PEAK-DT age-referenced scores, the minimum age at which a typically developing peer would likely achieve the same PEAK-DT score was reported. For example, if a participant had a Foundational Learning Skills PEAK score of 33, then his or her age-referenced score would be 3 years based on the PEAK age-referenced table. Because the age-referenced score reports the minimum possible age, the age-referenced scores were reported in intervals of 2 years (i.e., 1,3,5,7,9). For each component, the possible scores varied based on when typically developing peers would likely achieve a maximum score in that component. For example, the age-referenced score for Perceptual Learning Skills ranged from 1 to 7, because a 1-year-old would be expected to have a minimum score of 0, and a 7-year-old would expect to achieve the maximum score for that component. Test–retest reliability was assessed using an ICC. ICCs are ideal for test-retest comparisons as they assess the rank of participant scores rather than the raw scores as a continuous variable.

To assess the interrater reliability of the assessment, a subset of 10 participants were reassessed by a second assessor. The second assessor was blind to the assessment results obtained by the first assessor. Interrater reliability of PEAK-DT was assessed through Cohen’s kappa

(κ), interobserver agreement of all items (IOA), and interobserver agreement of scored items (IOA-S). Cohen’s kappa was determined by comparing the score attributed to each item, for each participant, by two observers. Each item was scored as a “0” if a skill was absent and a “1” if the skill was present. Cohen’s kappa was used to eliminate biases in agreement that may have occurred by chance. IOA was assessed by determining the percentage of items where two observers agreed on the score. IOA-S was conducted by determining the percentage of items where two observers agreed, using only the items where at least one observer scored the item as a “1.” Using scored IOA reduces the bias that may result in high IOA, when several of the items are scored as absent from the repertoire of participants.

Results

The results of the PEAK-DT assessment suggest that the mean PEAK-DT score across all participants was 92.45 ($s = 80.17$) at Time 1 and 95.7 ($s = 79.04$) at Time 2, and the mean age-referenced PEAK-DT score was 4.08 ($s = 3.48$) at Time 1 and 4.18 ($s = 3.47$) at Time 2. The results of the reliability analyses conducted in the study are summarized in Table 2. The ICC for raw PEAK scores suggested a strong relationship ($ICC = .987, p < .001$) between PEAK-DT scores at the initial and second assessment times across participants. The results were consistent across PEAK-DT components, including Foundational Learning ($ICC = .940, p < .001$), Perceptual Learning ($ICC = .972, p < .001$),

Table 2
Test-Retest and Interrater Reliability of the PEAK-Direct Training Assessment

PEAK Factor	ICC (raw)	ICC (age)	κ	IOA	IOA-S
Foundational Learning	.940**	.887**			
Perceptual Learning	.972**	.937**			
Verbal Comprehension	.993**	.917**			
Verbal Reasoning, Memory, and Mathematical Skills	.988**	.953**			
Total score	.987**	.973**	.981**	.991	.986

Note. ICC (raw) = intraclass reliability coefficient based on raw PEAK score; ICC (age) = intraclass reliability coefficient based on age-referenced PEAK score; κ = Cohen’s kappa reliability coefficient; IOA = interobserver agreement; IOA-S = interobserver agreement of scored items.

** $p < .001$.

Verbal Comprehension ($ICC = .993, p < .001$), and Verbal Reasoning, Memory, and Mathematical Skills ($ICC = .988, p < .001$). In addition, the ICC for age-references PEAK scores are also displayed in Table 2. There was a strong relationship for PEAK –DT Total score at Time 1 and Time 2 (.973), and this was consistent across PEAK-DT components, including Foundational Learning (.887), Perceptual Learning (.937), Verbal Comprehension (.917), and Verbal Reasoning, Memory, and Mathematical Skills (.953). Table 2 also displays results of the interrater reliability analyses. Cohen's kappa suggests that there was a strong relationship between the scores of the independent observers ($\kappa = .981, p < .001$), and both total and scored IOA suggested a high percentage of agreement between observers (99% and 98%, respectively).

Discussion

With the increased need for treatments of individuals with language deficits, ABA-based assessments are becoming increasingly necessary in order to identify specific skill deficits in order to guide treatment decisions. One such assessment is the PEAK-DT assessment. The present study provided an initial analysis of the reliability of the PEAK-DT assessment. The data suggest that PEAK-DT had high test-retest and interrater reliability. Results further indicated that strong test-retest correlations were observed across all four components of the PEAK-DT module, both in terms of raw scores and age-referenced scores. Close comparison of the ICCs obtained for PEAK-DT raw and age-referenced scores suggest that the raw scores produced higher correlations than age-referenced scores across all components. As such, using raw scores to report PEAK-DT performance may provide a more reliable measure. However, there may be social validity to using age-referenced scores instead of raw scores, and the correlations obtained for the age-referenced scores were also high. These results both replicate and extend previous research by indicating that the PEAK-DT assessment is reliable. The data suggest that the PEAK-DT can be administered across time and by different assessors while still providing reliable results. PEAK provides a way to target skills that are absent and may remain absent without specific targeted training for up to 1

month. The utility of the present study, beyond demonstrating the reliability of the PEAK-DT assessment, may also serve to facilitate interest from other disciplines who may be more familiar with psychometric evaluations of assessment tools.

A possible limitation of the present study is the sample size. The limited sample size reduces the external validity of the results obtained in the present study. Despite this limitation, the significance of the data is important because significance is less likely with smaller sample sizes. With small sample sizes each individual score has greater influence on the overall data. A second limitation is that the reliability of only one PEAK module, the PEAK-DT, was tested. With there being a total of four modules in the PEAK protocol, further research is necessary to assess the validity and reliability of the remaining PEAK modules. A third limitation was that differences observed between items that were directly tested and those that were not was not assessed, which would have provided an additional measure of validation. A final limitation is that only participants with disabilities took part in the study. This limitation restricts the inferences of reliability to PEAK-DT assessments that are conducted with individuals with disabilities, especially autism. Although the reliability of the assessment with other populations may be inferred, more research is needed to evaluate the reliability of the assessment with other populations.

Beyond addressing the limitations of the current study, there are several avenues that future research may explore to increase an understanding of the psychometric properties of PEAK-DT. First, future research may continue to explore other psychometric properties of the PEAK-DT assessment. One potential avenue is assessing the content validity of PEAK-DT as an assessment of directly trained language. Although PEAK-DT has demonstrated correlations with other language-based assessments, the totality of the assessment within the confines of our understanding of language as a unified construct has not been addressed. Second, the assessments conducted in the present study were completed by highly trained behavior analysts. High reliability scores were obtained, however the degree to which the same scores would have been obtained by staff who were less trained or had less familiarity with the PEAK assessment

is unknown. Currently, there are no studies suggesting the training dosage that would be required to accurately conduct PEAK assessments, providing another avenue for future studies. In addition, the degree to which staff who have not established previous familiarity with participants prior to conducting the assessments has not been assessed. Future research may explore these questions to extend the external validity of the present findings.

Conclusions

The PEAK-DT is a recently developed assessment that utilizes discrete trial training to teach language deficits experienced by those with intellectual or developmental delays. Data from the current study provide an initial evaluation of the reliability for the PEAK-DT assessment, and the PEAK age distribution table. The results indicated that both the PEAK-DT raw score and age-referenced scores had high test-retest and interrater reliability, for both total scores and for each of the components. The data from the present study, in combination with the data from previous research (Dixon et al., 2015; Dixon, Belisle, Whiting, et al., 2014; Rowsey et al., 2015) support the PEAK-DT as both a reliable and valid assessment.

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(Appendix follows)

Appendix

PEAK Age-Referencing Table

PEAK Relational Training System: Direct Training Module

PEAK factor	Student score	Typical age score	Difference score	Approximate age equivalent
Foundational Learning Skills				
Perceptual Learning Skills				
Verbal Comprehension Skills				
Verbal Reasoning, Memory, and Mathematical Skills				
Total score				

Typical Age Distribution of PEAK Factor Scores*

PEAK factor	1–2 yrs	3–4 yrs	5–6 yrs	7–8 yrs	9–10 yrs
Foundational Learning Skills	2	30	34	34	34
Perceptual Learning Skills	0	18	21	22	22
Verbal Comprehension Skills	0	19	80	94	100
Verbal Reasoning, Memory, and Mathematical Skills	0	0	10	22	28
Total score	2	67	141	172	184

* Items within abilities of >80% of typical developing children as reported by: [Dixon, Belisle, Whiting, & Rowsey \(2014\)](#).

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